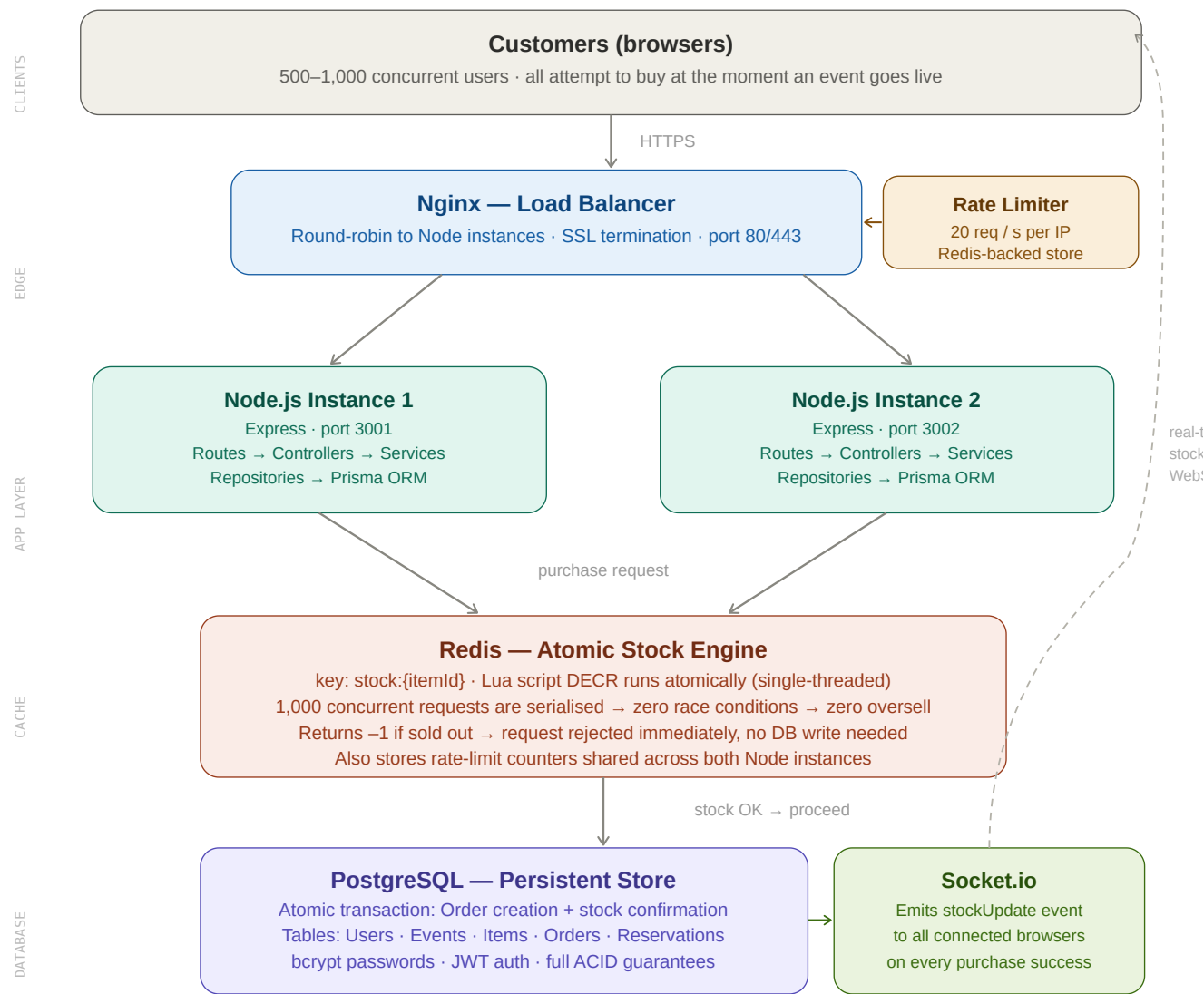


SwiftDrop — High-Concurrency Architecture

Full-Stack Engineering Challenge · Bitcode V6.0 · bitcode-swiftdrop



How concurrency is solved — judge walkthrough

- Redis Lua script is atomic — 1,000 simultaneous Buy requests are serialised one-by-one. Oversell is impossible.
- Nginx round-robin across 2 Node instances — no single server is overwhelmed. If one crashes, traffic continues. (NFR02)
- Rate limiter (20 req/s per IP, Redis-backed, shared across instances) blocks burst abuse, prevents 5xx crashes. (NFR03)
- PostgreSQL atomic transaction wraps Order creation + stock decrement — no order without a stock unit. (NFR07)
- Socket.io pushes live stock counts to all browsers via WebSocket — no polling, instant sold-out display. (FRM03, FRM04)
- Clean architecture: Routes → Controllers → Services → Repositories. Zero business logic in route handlers. (NFR08)